

Statistics for nuclear and particle physicists

LOUIS LYONS



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‘The result of this experiment was inconclusive,
so we had to use statistics.’

(Overheard at international physics conference)

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PREFACE

The material in this book is an enlarged version of a highly successful, short course of lectures given to graduate students in the Nuclear Physics Laboratory, Oxford. The course was designed to interest both nuclear structure and elementary particle physicists.

I am an experimental high energy physicist, and although as an undergraduate I suffered a course in statistics, it was only later during my research work, when I had to deduce something from my own data, that I learned how to use statistics. I also discovered that there were many things that one learned by experience and which were not explicitly mentioned in text-books; I tried to incorporate these aspects of the subject in my lectures. The emphasis of my lectures was thus not on formal proofs or on a rigorous treatment of the subject, but rather on practical applications and on how to use statistics to obtain the best results from one's data and to know the limitations of the results. In short, this was a course given by a non-statistician to non-statisticians.

My course did not attempt to cover every single example of statistics problems that can arise in nuclear and particle physics. The aim was rather to explain as fully as possible the different techniques that are available for attacking data analysis problems, to explain their relative merits and drawbacks, and to try to give the students sufficient confidence in their own ability to tackle any new problems that they might encounter. The book has maintained this approach.

There is an old joke about a man who was asked whether he could play the violin, to which he replied that he did not know as he had never tried. Instead of being about violin-playing, this story could equally well have been about the application of statistics. I firmly believe that one cannot learn statistics simply by reading a book on the subject. I have thus included a short series of problems which every reader is strongly recommended to complete. The majority of these were set for the Oxford graduate students and used to assess their progress. They claim that the time taken to do these problems was $\sim 40 \pm 10$ hours.

I am grateful to John Lloyd for his valuable comments on a preliminary draft of this book, to many colleagues with whom I have had discussions over the years on statistics, and to all the students whose questions and comments have shown me which parts of the course were in need of further clarification. Finally I want to thank Prof. Yehuda Eisenberg for extending the hospitality of the Physics Department of the Weizmann Institute; the main draft of this book was written in its congenial atmosphere.

L. Lyons, 1985

Experimental errors

1.1 Why do we do experiments?

There are basically two different types of results of experiments that scientists perform in order to learn about the physical world. In one type, we set out to determine the numerical value of some physical quantity, while in the second we are testing whether a particular theory is consistent with our data. These two types are referred to as ‘parameter determination’ and ‘hypothesis testing’ respectively. (Of course, in real life situations there is a degree of overlap between the two: a parameter determination may well involve the assumption that a specific theory is correct, while a particular theory may predict the value of a parameter.) For example, a parameter determination experiment could consist of measuring the velocity of light, while a hypothesis testing experiment could check whether the velocity of light has suddenly increased by several percent since the beginning of this year.

In this chapter, we are mainly concerned with various aspects of calculating the accuracy of parameter determination type experiments. We will have more to say about hypothesis testing experiments in Chapter 2.

1.2 Why estimate errors?

When we performed parameter determination experiments at school, we considered that the job was over once we obtained a numerical value for the quantity we were trying to measure. At university, and even more so in every-day situations in the laboratory, we are concerned not only with the answer but also with its accuracy. This accuracy is expressed by quoting an experimental error on the quantity of interest. Thus a determination of the velocity of light might yield an answer

$$c = (3.09 \pm 0.15) \times 10^8 \text{ metres/sec.}$$

In Section 1.4, we will say more specifically what we mean by the error

of ± 0.15 . At this stage it is sufficient to say that the more accurate the experiment the smaller the error; and that the numerical value of the error gives an indication of how far from the true answer this particular experiment may be.

The reason we are so insistent on every parameter determination including an error estimate is as follows. Scientists are rarely interested in parameter determination for its own sake, but more often will use it to test a theory, to compare with other experiments measuring the same quantity, to use this parameter to help predict the result of a different experiment, and so on. Then the numerical value of the error becomes crucial in the interpretation of the result.

For example, let us return to our determination of the velocity of light, and let us use the result to see whether it does indeed look as if its value has increased by, say, a few percent compared with its previously accepted value of 2.998×10^8 metres/sec.†

With a new measurement of 3.09×10^8 metres/sec, do we have evidence for an increase? There are essentially three possibilities.

Possibility 1

If the experimental error is ± 0.15 , then the new determination looks perfectly consistent with the old one, and we have no grounds for any new unconventional theory,
i.e. 3.09 ± 0.15 is consistent with 2.998.

Possibility 2

If the experimental error is ± 0.01 , then the new determination is inconsistent with the previously accepted value. Hence, on the basis of this new experiment, there does appear to be evidence for the idea that the value of c has increased (or that our experimental result and/or the error estimate are wrong),
i.e. 3.09 ± 0.01 is inconsistent with 2.998.

Possibility 3

With an error of ± 2 , the two values of c are consistent but the accuracy of the new experiment is so low that it would be incapable of detecting a change in c of even much more than a few percent,
i.e. 3.09 ± 2 is consistent with 2.998, and with many other values too.

† Since this is an experimental number, it too has an experimental uncertainty, but c has been measured so well that the uncertainty arises in the eighth figure after the decimal point, so we can effectively forget about it here.

Thus for a given value of the parameter, our reaction – ‘Conventional physics is in good shape’ OR ‘We have made a world shattering discovery’ OR ‘We should find out how to do better experiments’ – in this case depends on the numerical estimate of the accuracy of our experiment. Conversely, if we know only that the result of the experiment is that the value of c was determined as 3.09×10^8 metres/sec (but do not know the value of the experimental error), then we are completely unable to judge the significance of this result.

The moral is clear. Whenever you determine a parameter, estimate the error or your experiment is useless.

1.3 Random and systematic errors

There are two fundamentally different sorts of errors associated with any measurement procedure, namely random and systematic errors. The random errors come simply from the inability of any measuring device to give infinitely accurate answers, while the systematic errors are more in the nature of mistakes.

These different types of error are illustrated by considering an experiment involving counters to determine the decay constant λ of a radioactive source. The necessary measurements consist of counting how many decays are observed in a given time in order to determine the decay rate ($-dn/dt$); and of weighing the sample (in order to obtain the number of nuclei n present). Then the decay constant λ is calculated from the formula

$$-\frac{dn}{dt} = \lambda n. \quad (1.1)$$

The main random error comes from the fact that there is an inherent statistical error involved in counting random events. Other contributions to the random error come from the timing of the period for which the decays are observed and from the uncertainty in the mass of the sample.

The more obvious possible sources of systematic error are:

- (i) The counters used for detecting the decays may not be completely efficient and/or they may not completely surround the source, and so the counting rate may be below the true decay rate.
- (ii) The counters may be sensitive to particles coming from other than the source (e.g. cosmic rays), which would give a counting rate above the true decay rate.
- (iii) The radioactive source may not be pure (either chemically or

- isotopically) and so the number of nuclei capable of decaying may be less than the number as deduced from the mass of the sample.
- (iv) There may also be calibration errors in the clock and balance used for measuring the time interval and the sample mass.

A recurring theme in this book is the necessity of providing error estimates on any measured quantity. Because of their nature, random errors will make themselves apparent by producing somewhat different values of the measured parameter in a series of repeated measurements. The estimated accuracy of the parameter can then be obtained from the spread in measured values as described in Section 1.4.2.

An alternative method of estimating the accuracy of the answer exists in cases where the spread of measurements arises because of the limited accuracy of measuring devices. The estimates of the uncertainties of such individual measurements can be combined as explained in Section 1.5 in order to derive the uncertainty of the final calculated parameter. This approach can be used in situations where a repeated set of measurements is not available to employ the method described in the previous paragraph. In cases where both approaches can be used, they should of course yield consistent answers.

The accuracy of our measurements will of course play little part in determining the accuracy of the final parameter in those situations in which the population on which the measurements are made exhibits its own natural spread of values. For example, the heights of ten-year-old children will in general be scattered by an amount which is larger than the uncertainty with which the height of any single child can be measured.

An intermediate situation arises where the observation consists in counting independent random events in a given interval. Although the spread of values will usually be larger than the accuracy of counting (which may well be exact), it is known (see Section 3.2) that for an expected number of observations n , the spread is $\sqrt{n}$.

For systematic errors, however, the 'repeated measurement' approach will not work; if our detector, which we believe counts all particles passing through it, in fact has an efficiency of only 1%, the decay rate will come out too small by a factor of ~ 100 each time we repeat the experiment, and yet everything will look consistent.

Systematic errors can arise on any of the actual measurements that are required in order to calculate the final answer. But they can also appear indirectly in variables which do not explicitly appear in any formulae. Thus, for example, maybe the derivation of the formula being used requires

all the apparatus to be at a constant temperature; or we really ought to be performing our experiment in a vacuum instead of in the laboratory atmosphere; or the presence of some impurity is producing spurious results. Ideally, of course, all such effects should be absent. But if it is known that such a distorting effect is present, than at least some attempt can be made to estimate its importance and to correct for it. In effect, we are then converting what was previously a systematic error into what is hopefully only a random one.

In general, there are no simple rules or prescriptions for eliminating systematic errors. To a large extent it is simply a matter of common sense plus experience to know what are the possible dangerous sources of errors of this type. But one possible check that can sometimes be helpful is to use constraints that may be relevant to the particular problem. We give three simple illustrations of this.

Example (a)

We want to know whether a certain protractor has been correctly calibrated. One possible test is to use this protractor to measure the sum of the angles of a triangle. We then repeat this for several triangles and plot a histogram of our results. If the resulting distribution peaks significantly away from 180° , our protractor may be in error.

Example (b)

If a radioactive source emits electrons and photons in coincidence, then (after we have corrected for the inefficiencies of the various counters) we should observe the same number of counts in our electron as in our photon detector. A similar check could be made for a particle whose branching ratios for decays to electrons and to muons are equal.

Example (c)

Nuclear interactions conserve energy and momentum. Thus if we observe a series of interactions and plot the various components of the momentum and the energy imbalance, we expect to obtain distributions which peak around zero. Otherwise there is evidence that something is wrong. For example, the geometrical locations of various pieces of apparatus may not be exactly as we thought, our measurements of the magnitude of the momentum may be systematically in error, there could be some undetected particle taking away significant momentum and energy, etc. (See Fig. 1.1.)

Thus in the above examples, the existence of constraints enables at least

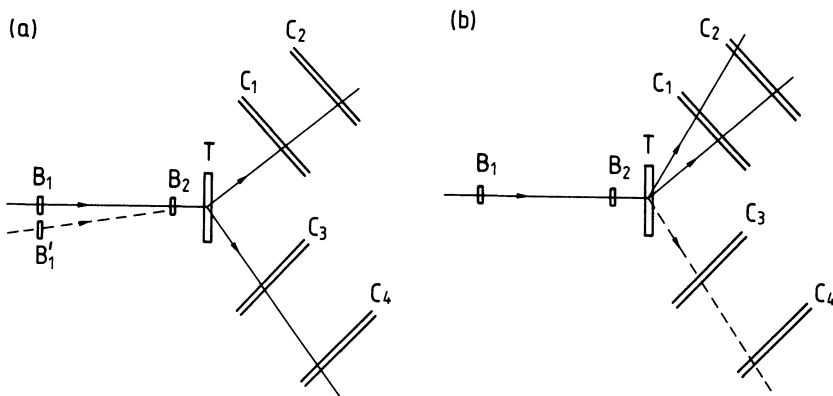


Fig. 1.1. Schematic diagram of apparatus to select examples of a reaction with two outgoing particles. T is the target; B are detectors for the beam particle and C for the outgoing secondaries. (a) Because of alignment problems, counter B_1 is incorrectly thought to be in the position B_1' . This results in the beam tracks' directions being systematically in error. This should become apparent by the failure of p_T , the momentum transverse to the beam, to balance. By moving B_1' , its correct position can be found as that which gives the best balance of p_T . An incorrect position for B_1 would give an incorrect value of, for example, the scattering angle θ . (b) In this example, two charged particles passing through C_1 and C_2 are detected, but a neutral particle (shown dashed) passing through C_3 and C_4 is not detected. The momenta of the three outgoing particles balance that of the beam, but with only two of them detected, the observed momenta appear not to balance. This failure to satisfy the momentum constraints enables us to reject this particular event as not being an example of the two body final state that we wish to study (but we clearly need to be careful not to reject events as in (a) above simply because we have made a small error in the assumed positions of the counters).

some check to be performed on the possible existence of systematic effects. In all cases, of course, random effects are also present, and so we do not expect the constraints to be identically satisfied in any particular case. The virtue of using a repeated series of observations is that the spread in the final histogram gives an indication of the magnitude of the random effects,[†] and hence we can then see whether there is any evidence for a systematic effect too. Random errors are usually more amenable to systematic study, and the rest of this chapter is largely devoted to them.

[†] The presence of a systematic effect of varying magnitude can broaden an observed distribution as well as shifting it from its expected position. Thus the observed width of an experimental distribution may give an over-estimate of the random effects.

Finally we assert that a good experimental physicist is one who minimises and realistically estimates the random errors of his apparatus, and who reduces the effect of systematic errors to a much smaller level.

1.4 The meaning of σ

In this section we are going to consider in more detail what is meant by the error σ on a measurement. However, since this is related to the concept of the spread of values obtained from a set of repeated measurements, whose distribution will often resemble a Gaussian distribution, we will first have three mathematical digressions into the subjects of (a) distributions in general, (b) the mean and variance of a distribution, and (c) the Gaussian distribution.

1.4.1 Distributions

A distribution $n(x)$ will describe how often a value of the variable x occurs in a defined sample. The variable x could be continuous or discrete, and

Table 1.1. *Examples of distributions*

Character	Limits	x variable	$n(x)$
Discrete	$1 \rightarrow \infty$	Integer x	No. of times you have produced a completely debugged computer program after x compilations
	$1 \rightarrow 7$	Day of week	No. of marriages on day x
	$-13.6 \text{ eV} \rightarrow 0$	Energy of ground and excited states of hydrogen atoms	No. of atoms with electrons in state of energy x in atomic hydrogen at 30 000°
Continuous	$-\infty \rightarrow \infty$	Measured value of parameter	No. of times measurement x is observed
	$0 \rightarrow \infty$	Time it takes to solve all problems in this book	No. of readers taking time x
	$0 \rightarrow 24 \text{ hours}$	Hours sleep each night	No. of people sleeping for time x

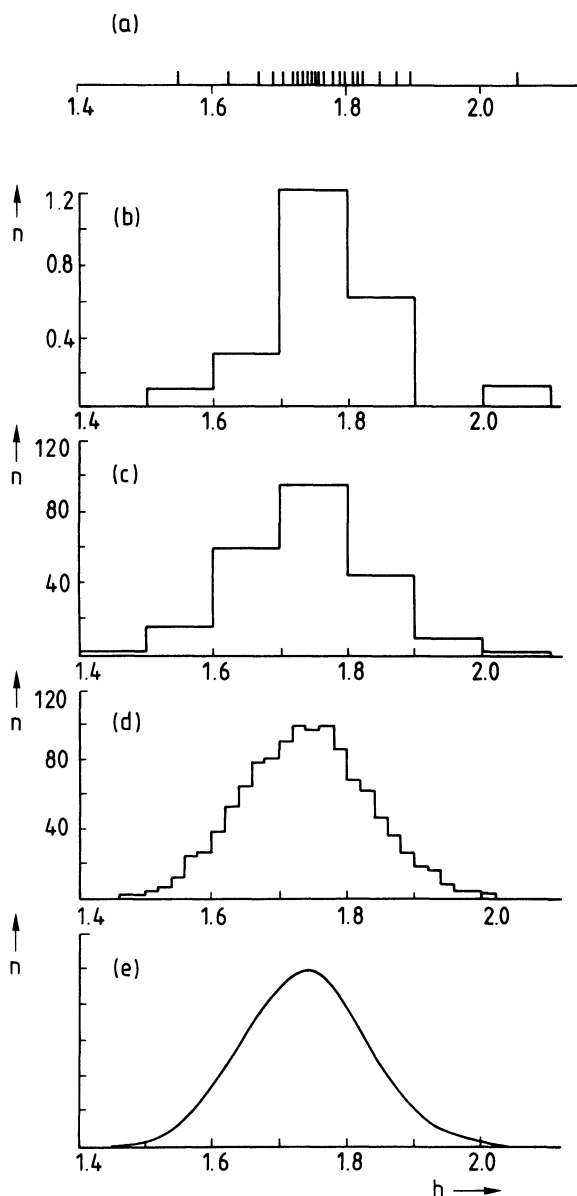


Fig. 1.2. Some examples of distributions of, say, the heights h (in metres) of 30-year-old men. (a) With only a few observations, each one is represented as a bar at the relevant position along the h -axis. (b) The data of (a) could alternatively be drawn as a histogram, where n is the number of men per cm interval of h , even though the bin size in h is 10 cm. (c) A histogram as in (b), but with 100 times more observations. (d) The same data as in (c), but drawn with smaller bins of h . n is still the number of men per cm interval of h . (e) For an even larger number of observations and with smaller bin size, the histogram of (d) approaches a continuous distribution.

its values could be confined to a finite range (e.g. 0–1) or could extend to $\pm \infty$ (or could occupy a semi-infinite range, e.g. positive values only). Some examples are given in Table 1.1.

Fig. 1.2(a) shows an example of a possible distribution of a continuous variable from a few observations of, say, the height h of 30-year-old men. Since only a few values are available, the data are presented by marking a bar along the h -axis for each measurement. In Fig. 1.2(b), the same data is presented as a histogram, where a fairly wide bin size for h is used and the vertical axis is labelled as n , the number of observations per cm interval of h , despite the fact that the bin size (Δh) used is 10 cm. The actual number of men in a given bin is $n \Delta h$, and the total number of men appearing in the histogram is $\sum n \Delta h$. If 100 times more measurements were available, the number of entries in each bin of the histogram would increase by a large factor (Fig. 1.2(c)), but it would now become sensible to draw the histogram with smaller bins, in order to display the shape of the distribution with better resolution. Because we plot $n(h)$ as the number of observations per cm, irrespective of bin size, the overall height of the histogram does not change much when we change the bin size (see Fig. 1.2(d)). Finally, for an even larger number of observations, we could make the bin size so small that the histogram would approximate to a continuous curve (see Fig. 1.2(e)); this could alternatively be viewed as a very good theoretical prediction about the numbers of men of different heights. Again $n(h)$ is to be interpreted in the same way, but now the total number of men appearing in the histogram is $\int n(h) dh$.

1.4.2 Mean and variance

In order to provide some sort of description of a distribution such as shown in Fig. 1.2, we need measures of the x -value at which the distribution is centred, and how wide the distribution is. The mean μ and the mean square deviation from the mean σ^2 (also known as the variance) are suitable for this. For a set of N separate measurements such as shown in Fig. 1.2(a), they are defined as†

$$\bar{x} = \sum x_i / N \quad (1.2)$$

$$\text{and} \quad s^2 = \sum (x_i - \mu)^2 / N. \quad (1.3)$$

† We adopt the convention that the true mean and variance are denoted by μ and σ^2 , while the measured mean and variance of a sample are $\bar{x}$ and s^2 .

The summations in eqns (1.2) and (1.3) extend over the N values of the sample. In general, the true mean μ is not known, and so eqn (1.3) cannot in fact be used to estimate the variance. Instead it is replaced by

$$s^2 = \frac{1}{N-1} \sum (x_i - \bar{x})^2. \quad (1.3')$$

Thus one measurement of a quantity does not allow us to estimate the spread in values, if the 'true' value is not known. (Other problems of using an estimated rather than a true value of a parameter appear in Sections 1.6 and 3.3.)

As an aside, we note that the factor $1/(N-1)$ in eqn (1.3') is required in order to make s^2 an *unbiased* estimate of the population's variance σ^2 , i.e. for a large sample, s^2 will tend to σ^2 . If, however, we rewrite the equation as

$$s^2 = \frac{1}{N+k} \sum (x_i - \bar{x})^2 \quad (1.3'')$$

and set $k = 0$ rather than -1 , although the estimate s^2 is now biased, it turns out that its mean square spread from the correct value σ^2 is smaller† than that for the unbiased estimate (1.3'). In fact, using $k = +1$ results in that estimate s^2 which has the smallest mean square spread from σ^2 . (See problems 5.12 and 6.4.) However, we regard $k = -1$ as preferable. This is because we are usually interested in using the variance in order to judge the significance of the deviation of a measurement $x \pm s$ from some preconceived value y . We then calculate $(x-y)^2/s^2$, which should be distributed like Student's t distribution, but especially if the details of how s^2 was obtained are unavailable, we will be satisfied in assuming that it is Gaussian distributed (see Section 3.3). In that case, it is irrelevant to have a minimum spread estimate for s^2 , since we are interested in $1/s^2$, and it is unreasonably small estimates of s^2 which would give trouble. It is because the choices $k = 0$ and $+1$ bias the estimates of s^2 downwards that we regard eqn (1.3') as preferable.

It is most important to realise that s is the measure of how spread out the distribution is, and is not the accuracy to which the mean $\bar{x}$ is determined. This is known to an accuracy better by a factor of $\sqrt{N}$. Thus

† The mean square deviation d_0^2 of a set of measurements from the true value t is related to the mean square deviation d^2 from their mean m by

$$d_0^2 = d^2 + b^2,$$

where b is the bias, i.e. $m - t$. As k increases, d^2 decreases but b^2 increases, and there is an optimum value of k which minimises d_0^2 .

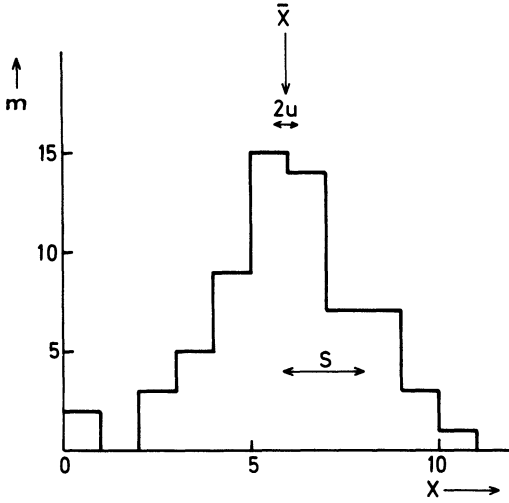


Fig. 1.3. A histogram of a distribution in x . m is the number of entries in each x bin; there are 66 entries in all. The mean $\bar{x}$ and the variance s^2 are estimated as 5.9 and $(2.05)^2$ respectively. The accuracy u to which the mean $\bar{x}$ is determined is smaller than s by a factor of $\sqrt{66}$.

by taking more and more observations of x , the variance s^2 will not change (apart from fluctuations) since the numerator and denominator of eqn (1.3) or (1.3') grow more or less proportionally; this is sensible since s^2 is supposed to be an estimate of the variance of the overall population, which is clearly independent on the sample size N . On the other hand, the variance of the mean (s^2/N) decreases with increasing N ; more data help locate the mean to higher accuracy. (We return to this point, and explain the origin of the $1/N$ factor, in Section 1.5.)

A minor computational point is worth noting. Formula (1.3') can be written

$$\begin{aligned} s^2 &= \frac{1}{N-1} \sum (x_i - \bar{x})^2 \\ &= \frac{N}{N-1} [\overline{x^2} - \bar{x}^2], \end{aligned} \quad (1.4)$$

where $\overline{x^2}$ is defined in analogy with eqn (1.2) as

$$\overline{x^2} = \sum x_i^2 / N. \quad (1.5)$$

Thus it is not necessary to loop over the data twice (first to calculate $\bar{x}$ and then to obtain s^2 from (1.3')), but $\overline{x^2}$ and $\bar{x}$ can be calculated in the same loop.

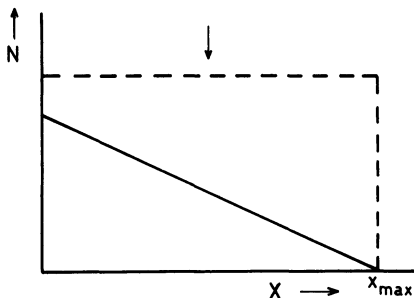


Fig. 1.4. The dashed line corresponds to a uniform distribution in x , with mean value denoted by the arrow. If the experimental efficiency for detecting events falls off linearly from 80% at $x = 0$ to zero at $x_{\max}$, the observed distribution will correspond to the solid line. In order to obtain the correct mean (or any other parameter), the observed distribution must first be corrected for the varying detection efficiency.

Sometimes the measurements are grouped together so that at the value x_j there are m_j events (equivalent to $n_j \Delta h$ in Fig. 1.2(d)). Then simple extensions of eqns (1.2) and (1.3') are

$$\bar{x} = \sum m_j x_j / \sum m_j \quad (1.6)$$

and

$$s^2 = \sum m_j (x_j - \bar{x})^2 / (\sum m_j - 1). \quad (1.7)$$

The variance on the mean u^2 is $s^2 / \sum m_j$ (see Fig. 1.3).

For the continuous distributions of Fig. 1.2(e), these become

$$\bar{x} = \int n(x) x dx / N \quad (1.8)$$

and

$$s^2 = \int n(x) (x - \bar{x})^2 dx / N, \quad (1.9)$$

where

$$N = \int n(x) dx,$$

and where the usual $N-1$ factor in s^2 has been replaced by N which is assumed to be large for this case.

Another possibility is that the individual measurements have been made with different detection efficiencies, perhaps because they were made with different parts of the apparatus or perhaps because the intrinsic efficiency is a function of the variable x (see Fig. 1.4). Then it is necessary to weight each measurement x_k by a factor w_k which is the reciprocal of the detection